Experimental Demonstration of Beam-Steered Optical Wireless Localization: A Single-LED Sensing Modality for 6G Integrated Sensing and Communication

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***Abstract*—Integrated sensing and communication (ISAC) is a defining feature of the ITU IMT-2030 (6G) vision, in which centimeter-level positioning becomes a native network service rather than an add-on. Optical wireless communication (OWC) is a strong candidate for the 6G optical access layer, and beam-steered transmitters—already investigated for high-capacity, mobility-robust 6G optical links—offer a largely untapped opportunity to deliver high-accuracy localization from the very same infrastructure. Building on our model-based beam-steered single light-emitting diode (LED), single-photodiode (PD) positioning framework, so far validated only in simulation, this paper reports its first experimental demonstration on an automated three-dimensional (3D) robotic testbed. A single near-infrared LED is steered through K predefined orientations while a PD held at an arbitrary, fixed orientation records the received-signal-strength (RSS) profile; the transmitter-to-receiver direction is recovered by ratio-based least-squares estimators that are provably independent of the receiver orientation, and the range is obtained either *cooperatively* (a single beam-aligned measurement) or in *broadcast* mode (from the same K measurements). We further introduce an empirical quasi-Lambertian radiation model, calibrated from a dedicated radiometric campaign, and embed it into the estimators to compensate for the angular asymmetries of commercial off-the-shelf LEDs. Over a densified grid of $\approx [2000]$ positions spanning a $[3.0 \times 3.0 \times 1.4]$ m workspace, the system attains a mean direction-finding error of $[X.X]^\circ$ and a 3D positioning error of $[X.X]$ cm (90th percentile $[X.X]$ cm), degrading by less than $[X]$% under receiver tilts up to $[30]^\circ$. These results experimentally confirm that beam-steered OWC can meet the IMT-2030 indoor positioning target with minimal receiver hardware, establishing it as a practical sensing pillar for 6G optical ISAC.**

***Index Terms*—6G, integrated sensing and communication (ISAC), optical wireless positioning, beam steering, LiFi, Cramér–Rao bound, receiver-orientation independence, radiation-pattern calibration, experimental demonstration.**

I. Introduction

[I-A. The 6G / ISAC opening — borrow the framing of [1] and [2]]. Open at the top of the wireless-systems hierarchy: while 5G is still being rolled out, standards bodies (ITU) have defined the IMT-2030 (6G) vision, in which integrated sensing and communication (ISAC) is a headline capability [2], [3]. State that the sensing function explicitly includes positioning, for which IMT-2030 targets an accuracy in the 1–10 cm range, and that localization is now recognized as a *key enabler* of 6G rather than a peripheral service [4], [5]. This paragraph reframes the whole contribution away from "just indoor 3D OWP" and toward "a sensing pillar for 6G."

[I-B. Why optical, and why the transmitter side]. Motivate the optical spectrum as one of the promising 6G bands, and OWC/LiFi as a mature dual-use medium where illumination, data, and positioning can coexist [6], [7]. Then introduce the central technological trend: 6G optical access is moving toward *transmitter-side spatial diversity and beam management*. Two complementary families illustrate this: (i) dense VCSEL/grid-of-beam arrays with adaptive divergence control for coverage and capacity [8], [9], and (ii) single-beam steering via MEMS micromirrors, liquid-crystal deflectors, or optical phased arrays [10]–[13]. Key rhetorical move: the same beam-steering / beam-management hardware being deployed for 6G optical *communication* can be re-used for *sensing*—this hardware reuse is exactly the ISAC premise. Note the contrast with the VCSEL-array line of work (e.g., [9]): those papers optimize divergence for coverage; we exploit the angular RSS diversity created by steering for localization.

[I-C. The single-LED OWP landscape and our beam-steered paradigm]. Condense the single-LED positioning survey argument [14]–[16]: single-source OWP reduces infrastructure but classically pushes complexity to the receiver (tilted-PD arrays, mechanically rotating PDs, IMU fusion) [17]–[22]. Contrast our paradigm: shift spatial diversity to the transmitter through K steered orientations, keeping the receiver a single static PD. Cite our own prior model-based results—direction finding + cooperative distance recovery [23], receiver-orientation independence and tilt robustness [24], and the broadcast (K-only) distance recovery / multi-user variant [25] (GLOBECOM). Use a near-infrared LED so the positioning link is decoupled from illumination standards [14].

[I-D. The gap this paper fills]. State the gap plainly: all of the above beam-steered OWP results are *simulation-based* and assume an *ideal Lambertian* source and *perfectly known* noise statistics. Two things are missing for 6G credibility: (i) a real hardware demonstration at room scale with ground truth, and (ii) a treatment of the real, quasi-Lambertian / asymmetric radiation pattern of COTS LEDs [26], which biases RSS-only estimators. This motivates an *experimental* paper with an empirically calibrated channel, aligned with related ISAC-oriented calibration work [27], [28].

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[I-E. Contributions]. Present as an itemized list (draft below).

- **First experimental demonstration** of beam-steered single-LED / single-PD 3D optical localization on an automated robotic testbed, with UR5-referenced ground truth over a densified grid of $\approx$ [2000] positions (vs. the $\sim$300-point conference proof-of-concept).
- **Empirical quasi-Lambertian radiation model** for COTS LEDs, calibrated from a dedicated radiometric campaign and embedded into the direction-finding and distance-recovery estimators to remove peripheral (angular) bias.
- **Experimental validation of the two distance-recovery regimes**: cooperative ($K{+}1$ beam-aligned) vs. broadcast (K-only), quantifying the accuracy/latency trade-off on real data and against the theoretical PEB/$\mathrm{PEB_B}$.
- **Experimental confirmation of receiver-orientation independence** of the direction-finding stage under controlled PD tilts, validating the $\mathbf{n}_r$-agnostic property predicted in [23], [24].
- **A 6G-ISAC positioning benchmark**: mapping the measured accuracy, robustness, and latency onto the IMT-2030 KPI, and outlining the integration path with an OWC data link.

[I-F. Explicit novelty vs. GLOBECOM and TCOM — REQUIRED paragraph]. Write one crisp paragraph: "Compared with our conference paper [25], which introduced the broadcast concept in simulation only, and with [23], which developed the model-based direction/position estimators and bounds in simulation, this paper contributes (i) the first hardware realization and measurement campaign; (ii) an empirically calibrated (non-ideal) radiation model absent from both; (iii) experimental head-to-head of cooperative vs. broadcast recovery; and (iv) an explicit repositioning of the architecture within the 6G ISAC / IMT-2030 framework."

[I-G. Paper organization]. Standard closing paragraph referencing Sections II–V.

II. Beam-Steered Optical Wireless Localization for 6G ISAC: Working Principles

A. System Model

Recap the geometry: ceiling-mounted steerable LED at $\mathbf{t} = [0, 0, H]^\mathsf{T}$ steered along $\{\mathbf{n}_{t,i}\}_{i=1}^{K}$; single PD at unknown $\mathbf{r}$ with orientation $\mathbf{n}_r$. State the Lambertian LOS RSS model, the irradiance/incidence cosines $\cos\phi_i = (\mathbf{n}_{t,i}\cdot\mathbf{d})/d$, $\cos\psi = -(\mathbf{n}_r\cdot\mathbf{d})/d$, the compact factorization $\mu_i = \eta\, Q_i^m$ with $\eta = C\cos\psi/d^2$, and the N-sample MVU mean and Gaussian noise model. Cite [23], [29]. Explicitly flag which quantities will later be replaced by empirical calibration (m, C, and the shape $\cos^m\phi \to R(\phi)$).

B. Direction Finding (Receiver-Orientation-Agnostic)

Summarize the three estimators from [23]: closed-form GLS and lightweight WLS (ratio-based, $3{\times}3$ eigenproblem) and iterative NLS (normalized Lambertian fit on $\mathbb{S}^2$). State the key property to be tested experimentally: all three cancel the common amplitude and are mathematically independent of $\mathbf{n}_r$ [23], [24].

C. Distance Recovery: Cooperative vs. Broadcast

Two regimes tested on the same dataset: (a) *Cooperative / active receiver*: LED steered toward the estimated direction and PD reoriented toward the LED (emulated by the UR5 end-effector) for a dedicated $(K{+}1)$-th measurement—this is the [23] TCOM pipeline. (b) *Broadcast / passive receiver*: the PD stays static and d is recovered in closed form from the same K measurements via the profiled amplitude estimator $\hat\eta = \sum_i \hat\mu_i \hat Q_i^m / \sum_i \hat Q_i^{2m}$ and $\hat d = \sqrt{C\,\widehat{\cos\psi}/\hat\eta}$—this is the [25] GLOBECOM pipeline, enabling simultaneous multi-user positioning. Note that broadcast recovery uses the recorded UR5 pose so a single dataset supports both pipelines with no extra measurement.

D. Empirical Radiation Modeling for Estimation Enhancement

Contrast the ideal $\cos^m(\phi)$ profile with the empirical, quasi-Lambertian / azimuthally-asymmetric pattern $R(\phi,\alpha)$ of COTS LEDs [26]. Propose the integration framework: replace Q_i^m by a calibrated response $\hat R(\phi_i)$ (spline look-up table or multi-Gaussian fit) inside GLS/WLS/NLS and inside the $\hat\eta$ estimator, and recompute the radiometric constant C under the calibrated profile. Discuss how this removes the spatial (peripheral) bias that a mismatched Lambertian order introduces in RSS-only estimation.

III. Experimental Framework and Implementation

A. 3D Robotic Testbed and Hardware Upgrades

Describe the testbed: steerable NIR LED on a two-axis motorized gimbal (Thorlabs PRM1Z8, 0.1° resolution) at the ceiling; PD carried by a UR5 arm acting as programmable ground truth. List the upgrades over the conference setup: *(i) Thermal management* — aluminum heat sink + star-PCB mount enabling nominal 1 A drive (vs. 750 mA), improving SNR; *(ii) Receiver front-end / FOV* — baseline narrow-FOV PD (BPX 61, $\approx 60^\circ$) vs. wide-FOV upgrade (S6967, $\approx 90^\circ$), and the corresponding reachable workspace (2.4×2.4 m vs. 3.0×3.0 m); *(iii) High-density grid* — 0.2 m resolution yielding up to $\approx$ [2000] positions across heights $h \in [\,[0.2, 1.4]\,]$ m. Include a hardware/parameter table (LED power, $\Phi_{1/2}$, A_{det}, FOV, TIA gain, DAQ rate).

B. Kinematics and Coordinate Transformations

Give the rigid-body transforms needed for ground truth: (a) *Transmitter kinematics* — rotation matrices of the two-axis gimbal mapping commanded $(\phi_{\mathrm{cmd}}, \mathrm{az}_{\mathrm{cmd}})$ to $\mathbf{n}_{t,i}$; (b) *Receiver kinematics* — mapping the desired PD pose in the global testbed frame to the UR5 base/tool frames, and extracting $\mathbf{n}_r$ from `pose_UR5` (with a note on sign/convention vs. the theoretical model). Document UR5 repeatability ($\pm$0.03 mm) as the ground-truth uncertainty.

C. Radiometric Calibration Protocol

Two calibration sub-datasets from `dataset_specification.md`: *Sub-dataset 1 — $R(\phi)$*: at fixed known d, sweep the irradiance angle $\phi \in [0°, 90°]$ (step 2–5°) over several azimuth cuts $(0/90/180/270°)$ to capture asymmetry; record raw $V_{\text{raw}}[n]$, dark $V_{\text{dark}}[n]$, driving current I_{LED}, temperatures. Fit $\hat{R}(\phi)$ (spline / multi-Gaussian). *Sub-dataset 2 — constant C*: PD under the LED (nadir/zenith) at known distances (0.4–1.2 m); report C under the calibrated $\hat{R}$ vs. the ideal-Lambertian C.

D. Spatial Data-Acquisition Protocol (Main Campaign)

Describe Sub-dataset 3: for each of the $\approx$ [2000] points, record ground truth $(x, y, z$, full `pose_UR5`, $\mathbf{n}_r)$; for each of the K=9 TCOM-optimized orientations record $\mathbf{n}_{t,i}$, raw/dark samples (N=1000 @ 1 kHz) and timestamps; plus the cooperative $(K+1)$-th beam-aligned measurement. Repeat the full $\{K, K+1\}$ scan $M \in [10, 30]$ times per point for empirical noise statistics ($\hat{\sigma}^2$), enabling direct comparison with the σ^2 assumed in [23], [25]. Emphasize that storing `pose_UR5` lets the *same* data feed both cooperative and broadcast pipelines.

E. Receiver-Tilt Robustness Protocol

Describe the deliberate-tilt sweep (axis D2): on a subset of [100–200] points, repeat the $\{K\}$ scan with the PD tilted to $\theta_{\text{tilt}} \in \{0, 6, 15, 30\}°$ over varied tilt azimuths, replicating the PEB-vs-tilt analysis of the broadcast paper. This isolates the experimental impact of physical attitude on both direction finding ($\mathbf{n}_r$-agnostic) and distance recovery ($\mathbf{n}_r$-dependent).

F. Dataset Structure and Reproducibility

Specify the storage schema (one row per `(point_id, orientation_id, repeat_id)`, separate tables for $(K+1)$ and for the calibration sub-datasets), retention of raw $V_{\text{raw}}[n]$ for $\hat{\sigma}^2$ recomputation, and session metadata for traceability. State the intention to release the dataset (open-data / reproducibility angle, valued for JLT and for the 6G community).

IV. Experimental Results and Discussion

A. Empirical Radiation Pattern Characterization

Present the measured $\hat{R}(\phi)$ (and azimuthal asymmetry) vs. the best-fit ideal $\cos^m \phi$; report the fitted effective m, the RMS model mismatch, and the corrected vs. ideal C. This is the empirical foundation used by the later capstone subsection.

B. Baseline Direction Finding and 3D Positioning

Report GLS/WLS/NLS direction error (mean, median, P_{90}) and full 3D error (RMSE, P_{90}, APE) over the dense grid, for cooperative and broadcast recovery, compared against the theoretical DEB / PEB / PEB_{B} from [23], [25]. Include spatial heatmaps and CDFs. Discuss where measured error departs from the bound (peripheral bias $\rightarrow$ motivates the next subsections).

C. Broadcast vs. Cooperative Distance Recovery

Head-to-head on identical data: accuracy gap and, crucially, the operational trade-off (broadcast = passive, simultaneous multi-user, no PD reorientation; cooperative = one extra aligned measurement, higher accuracy). Report empirical latency/robustness. Tie back to the multi-user 6G use case (multi-robot / AGV / IoT).

D. Robustness to Receiver Tilt ($\mathbf{n}_r$-Agnosticism)

Use the tilt-sweep data to show experimentally that direction finding is (near-)invariant to θ_{tilt} while broadcast distance recovery degrades in the predicted way; quantify degradation vs. θ_{tilt} and vs. K. This validates the theoretical $\mathbf{n}_r$-independence and the tilt sensitivity of the amplitude term.

E. Performance Enhancement via Empirical Radiation Modeling

Compare baseline (ideal-Lambertian) estimation against the estimation that plugs in the calibrated $\hat{R}(\phi)$. Show the reduction in peripheral direction error and 3D RMSE; report the percentage improvement and the fraction of positions moved below the 1–10 cm IMT-2030 band. This is the headline experimental contribution.

F. Discussion: Implications for 6G Optical ISAC

Map results onto IMT-2030 KPIs (accuracy 1–10 cm, latency), argue the hardware-reuse ISAC synergy (same steered transmitter serves data + localization), and honestly scope the claim: this paper demonstrates the *sensing/localization pillar*; joint data transmission is the natural next integration step (not demonstrated here). Contrast positioning-oriented steering with coverage-oriented VCSEL divergence control [8], [9].

V. Conclusion and Future Work

Summarize: first experimental demonstration of beam-steered single-LED / single-PD 3D optical localization, empirically-calibrated radiation modeling, cooperative-vs-broadcast validation, and tilt robustness—together meeting the IMT-2030 centimeter-level target with minimal receiver hardware, positioned as the sensing pillar of 6G optical ISAC. Future work: (i) true joint communication + localization on the same steered link (full ISAC, e.g., m-CAP data on the NIR carrier [1]); (ii) non-mechanical / high-speed steering (liquid-crystal, OPA) for mobility [10], [11]; (iii) IMU-orientation-error mismatch analysis for broadcast mode; (iv) scaling to multi-AP and VCSEL-array transmitters [8], [9].

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